

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400197
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Arroyo; Yadhira Haydee et al.

Gesture recognition for advanced security

Abstract

Security can be provided for a user in a banking environment by detecting a gesture. For example, a system described herein can include a sensing device positionable to detect the gesture from the user. The system can include a processor and a non-transitory computer-readable medium that includes instructions executable by the processor to perform operations. The operations can include receiving a notification of the gesture from the sensing device. The operations can also include receiving an input from the user to select an automated teller machine (ATM) operation. The operations can further include determining an intent of the gesture from the sensing device. The operations can include controlling the ATM operation based on the input from the user. Additionally, the operations can include transmitting a request separate from the ATM operation based on the notification of the gesture.

Inventors: Arroyo; Yadhira Haydee (Chicago, IL), Hepford; Ryan (Raleigh, NC), Sinha; Gaurav (Hawthorne, CA)

Applicant: Truist Bank (Charlotte, NC)

Family ID: 1000008780910

Assignee: Truist Bank (Charlotte, NC)

Appl. No.: 17/848032

Filed: June 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230419280 A1	Dec. 28, 2023

Publication Classification

Int. Cl.: **G06Q20/40** (20120101); **G06Q20/10** (20120101); **G06V40/50** (20220101); **H04L9/40** (20220101)

U.S. Cl.:

CPC **G06Q20/1085** (20130101); **H04L63/083** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5519669	12/1995	Ross	340/541	G01S 15/02
5712830	12/1997	Ross	340/541	G01S 15/02
8630820	12/2013	Amis	702/186	G08B 25/016
10990955	12/2020	Skelsey	N/A	G06Q 20/3221
11087109	12/2020	Strohmann	N/A	G06V 40/1306
11216149	12/2021	Yoganandan	N/A	G06F 3/04886
11275446	12/2021	Franklin	N/A	G06F 9/453
11348357	12/2021	Tiwari	N/A	G06F 3/0416
11823536	12/2022	Arroyo	N/A	G06Q 20/4016
11935090	12/2023	Goetz	N/A	G06Q 30/0238
2011/0046920	12/2010	Amis	709/217	G08B 25/016
2014/0071037	12/2013	Cohen	345/156	G06V 40/25
2014/0118140	12/2013	Amis	340/539.13	G08B 25/08
2015/0294432	12/2014	Jones	705/325	G06Q 50/265
2016/0148488	12/2015	Tijerina	705/325	G07F 9/026
2016/0149945	12/2015	Tijerina	726/25	H04L 63/1433
2016/0277397	12/2015	Watanabe	N/A	H04L 63/0861
2018/0103859	12/2017	Provenzano	N/A	A61B 5/0024
2018/0137263	12/2017	Kurian	N/A	G06F 3/167
2018/0137264	12/2017	Kurian	N/A	H04L 67/12

2018/0292907	12/2017	Katz	N/A	G06F 3/017
2019/0094967	12/2018	Bisbee	N/A	G06F 3/016
2019/0312966	12/2018	Shin	N/A	H04W 4/90
2019/0333076	12/2018	Chen	N/A	G06V 40/172
2020/0084607	12/2019	Geller	N/A	H04W 4/90
2020/0413202	12/2019	Kusano	N/A	H04R 31/00
2021/0073511	12/2020	Buchan	N/A	G01N 29/2437
2021/0350099	12/2020	Buchan	N/A	H04L 9/3231
2022/0014916	12/2021	Gillis	N/A	H04W 12/30
2022/0198173	12/2021	Azumi	N/A	G06V 40/1306
2022/0261083	12/2021	Franklin	N/A	G06F 9/453
2023/0085885	12/2022	Sreeram	455/404.1	G06F 3/048
2024/0037529	12/2023	Lal	N/A	G06Q 20/229

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3033315	12/2017	CA	G08B 21/0453
3125055	12/2019	CA	G06K 9/00268
106934955	12/2016	CN	N/A
206584461	12/2016	CN	N/A
109377682	12/2018	CN	G06K 9/00228
113792624	12/2020	CN	N/A
WO-2015023109	12/2014	WO	G06F 15/16
WO-2016085827	12/2015	WO	G06Q 20/206
WO-2017035825	12/2016	WO	N/A

OTHER PUBLICATIONS

CN-113792624-A. English Language Translation. Google Translate. “Early warning security monitoring method for bank ATM”. 2018. <https://patents.google.com/patent/CN113792624A/en?q=CN-113792624-A> (Year: 2021). cited by examiner

CN-109377682-A. English Language Translation. Google Translate. “A kind of facial pose anomalous identification alarm ATM cash dispenser device”. 2018. <https://patents.google.com/patent/CN109377682A/en?q=CN-109377682-A> (Year: 2018). cited by examiner

WO-2017035825-A1 Google Patents English Translation. <https://patents.google.com/patent/WO2017035825A1/en?q=WO2017035825A1> (Year: 2024). cited by examiner

CA-3033315-A1 Google Patents English Translation. <https://patents.google.com/patent/CA3033315A1/en?q=CA+3033315+A1> (Year: 2024). cited by examiner

Google Patents English translation of CN-206584461-U. <https://patents.google.com/patent/CN206584461U/en?q=CN-206584461-U> (Year: 2024). cited by

examiner

Google Patents English translation of CN-106934955-A.

<https://patents.google.com/patent/CN106934955A/en?q=CN-106934955-A> (Year: 2024). cited by examiner

Google Patents English Language Translation of WO-2015023109-A1.

<https://patents.google.com/patent/WO2015023109A1/en?q=WO-2015023109-A1> (Year: 2024). cited by examiner

Primary Examiner: Sharon; Ayal I.

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to banking operations and, more particularly (although not necessarily exclusively), to gesture recognition for advanced security.

BACKGROUND

(2) Security can be a concern in banking environments such as an environment that includes an automated teller machine (ATM). A would-be thief can know that an individual using an ATM can leave the ATM with cash in their possession after a withdrawal of funds. If an individual is the victim of an ATM robbery, other personal property can be taken such as wallets, credit cards, jewelry, and personal information. There can be a need for security in banking environments.

SUMMARY

(3) Security can be provided for a user in a banking environment by detecting a gesture. For example, a system described herein can include a sensing device positionable to detect the gesture from the user. The system can include a processor and a memory that includes instructions executable by the processor device to perform operations. The operations can include receiving a notification of the gesture from the sensing device. The operations can also include receiving an input from the user to select an automated teller machine (ATM) operation. Additionally, the operations can include determining an intent of the gesture from the sensing device. The operations can include controlling the ATM operation based on the input from the user and transmitting a request separate from the ATM operation based on the notification of the gesture.

(4) In another example, a method described herein can include receiving a notification of a gesture from a sensing device. The method can further include receiving an input from a user to select an automated teller machine (ATM) operation. Additionally, the method can include determining intent of the gesture from the sensing device. Further, the method can include controlling the ATM operation based on the input from the user and transmitting a request to authorities based on the notification of the gesture.

(5) In an example, a non-transitory computer-readable medium includes instructions that are executable by a processor for causing the processor to perform operations including receiving a notification of a gesture from a sensing device. The operations can further include receiving an input from a user to select an automated teller machine (ATM) operation. Additionally, the operations can include determining intent of the gesture from the sensing device. Further, the operations can include controlling the ATM operation based on the input from the user. The operations can also include transmitting a request to authorities based on the notification of the gesture.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic of a banking environment in which a user can be provided advanced security based on gesture recognition according to one example of the present disclosure.
- (2) FIG. 2 is a flowchart of a process for providing advanced security based on gesture recognition according to one example of the present disclosure.
- (3) FIG. 3 is a block diagram of an example of a computing device for providing advanced security based on gesture recognition according to one example of the present disclosure.

DETAILED DESCRIPTION

- (4) Certain aspects and examples of the present disclosure relate to gesture recognition for advanced security. A user in a banking environment can initiate a protocol that can provide security for the user by making a gesture. For example, a user can arrive at an ATM to drop off money for business. The user can notice a suspicious person standing too close or otherwise acting suspiciously.
- (5) The user can respond to such a situation by making a gesture at a sensing device of the ATM or a wearable device, providing an input code for the ATM, or a combination thereof. The input can include typing a fake personal identification number (PIN) or a passcode. For example, the gesture can include a facial expression (e.g., a wink) toward a camera of the ATM or the user clenching their fist. A sensing device, such as a smartwatch worn by the user, can detect some of these gestures and recognize an intent of the gesture. For example, the intent of the gesture can be a distress signal. The fake PIN or passcode can be associated with distress and can be used to confirm the intent of the gesture. The intent of the gesture can be determined in such a way that the suspicious person is not alerted that the user is responding to them.
- (6) A computing device can receive notification of the gesture and the intent of the gesture from the sensing device. The computing device can control the ATM based on the input. For example, the computing device can lock the user's bank account based on the input (e.g., the ATM will not output cash in response to the user's input). The computing device can contact authorities to report the distress signal. In some examples, in response to the notification of the gesture, the computing device can cause the user's smartwatch, smartphone, and the ATM to record video using all available cameras. The authorities can arrive and stop any attempted criminal activity.
- (7) Gesture recognition can provide users with security in banking environments. By using a simple gesture, users can transmit a distress signal to a computing device and stop a crime without alarming a would-be criminal.
- (8) Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.
- (9) FIG. 1 is a schematic of a banking environment **100** in which a user **112** can be provided advanced security based on gesture recognition according to one example of the present disclosure. Included in the banking environment **100** are the user **112**, one or more sensing devices **130**, one or more communication networks **140**, an ATM **134**, and a computing device **132**. The banking environment **100** can also include authorities **136**. The one or more sensing devices **130** and the ATM **134** may send or receive communication with the computing device **132** over the one or more communication networks **140**. The one or more communication networks **140** may correspond to one or more Wide Area Networks (“WANs”), such as the Internet, through which the one or more sensing devices **130**, the ATM **134**, and the computing device **132** may communicate with servers via web browsers or client-side applications, to establish communication sessions, request and

receive web-based resources, and access other features of applications or services. Although illustrated separate from the ATM **134** in the banking environment **100**, in certain examples, the computing device **132** can be included within the ATM **134** and in other examples, the computing device **132** can be situated in a remote location away from the banking environment **100**.

(10) The one or more sensing devices **130**, which can include suitable sensor devices for accessing web-based resources or application-based resources, can be capable of accessing and establishing communication sessions with the computing device **132** through the one or more communication networks **140**. As illustrated in FIG. 1, sensing devices **130a-130c** correspond to mobile devices, including tablet computers **130a**, smartphones **130b**, and smart watches **130c**, which may access the computing device **132** via a Local Area Network (“LAN”) or Wide Area Network (“WAN”), as well as mobile telecommunication networks, short-range wireless networks, or various other communication network types (e.g., cable or satellite networks). Sensing devices **130d-130f** correspond to sensing devices associated with the ATM, including an ATM camera **130d**, an ATM microphone **130e**, and an ATM keypad **130f**. Although certain examples herein are described in terms of mobile devices, in other examples, the one or more sensing devices **130** may additionally or alternatively include other mobile or non-mobile devices (e.g., desktop computers, laptop computers, and the like) capable of accessing the computing device **132** via the one or more communications networks **140**. The one or more sensing devices **130** and the ATM **134** can be capable of recording audio or video of an ATM operation.

(11) The one or more sensing devices **130** can detect a gesture from the user **112**. The ATM **134** can receive input from the user **112**. The input to the ATM **134** can include a user's PIN, a passcode, or a fake PIN. The computing device **132** can receive notification of the gesture and an intent of the gesture from the one or more sensing devices **130**. In some examples, the input to the ATM **134** can verify the intent of the gesture from the user **112**. For example, a pre-determined code that is not a normal ATM code for the user **112** may confirm the gesture intent. The computing device **132** can receive the input from the user from the ATM **134** and can control ATM operations based on the input.

(12) In an example, the computing device can transmit a request to the authorities **136**. The authorities **136** can include a bank manager, police station, paramedics, an emergency call center, etc., or any suitable combination of the foregoing. In some examples, the computing device **132** can identify a police station located near the ATM **134**. Transmitting the request can include contacting the police station to report a distress signal. Additionally, the request can include informing a bank manager of a bank associated with the ATM **134** of the distress signal.

(13) FIG. 2 is a flowchart of a process **200** for providing advanced security based on gesture recognition according to one example of the present disclosure. Operations of flowcharts may be performed by software, firmware, hardware, or a combination thereof. The operations of the flowchart start at block **202**.

(14) At block **202**, the process **200** involves detecting, by a sensing device **130**, a gesture from a user **112**. The sensing device **130** can, for example, include a smart device, smart phone, a tablet computer, wearable technology like a smartwatch, a smart camera, a laptop computer, an ATM camera, an ATM microphone, an ATM keypad, a hidden input device (e.g., a hidden token device), a bank tablet/“kiosk”, or any suitable combination of the foregoing. There can be more than one sensing device **130** present.

(15) The gesture from the user **112** can include any form of movement or vocal utterance detectable by the sensing device **130**. For example, the user **112** can clench one or both of the user's fists to indicate a gesture. Gestures can also include voice codes such as a particular phrase, facial expressions (e.g., a wink), etc. Gestures can also include a motion of an extremity such as the user **112** stretching their legs, swinging their arms, etc. In some examples, a passcode entered into the ATM **134** can denote a gesture.

(16) At block **204**, the process **200** involves receiving a notification of the gesture from the sensing

device **130**. The sensing device **130** can be communicatively coupled to a computing device **132**. The sensing device **130** may send or receive communication with the computing device **132** over one or more communication networks **140**. The sensing device **130** can notify the computing device **132** of a sensed gesture from the user **112**. The notification can include a location of the user **112**.

(17) At block **206**, the process **200** involves receiving an input from the user **112** to select an ATM **134** operation. In some examples, the input can be a passcode. The passcode can verify an intent of the gesture from the user **112**. For example, when the intent of the gesture is a distress signal, the user **112** can also input the passcode into the ATM **134**. The passcode can confirm that the user **112** is in distress. In some examples, the passcode can contradict the gesture. For example, the user can make a gesture that indicates distress and then determine that they are not in distress (e.g., they may realize that the suspicious person is a friend after making the gesture). The user can input a passcode that contradicts the intent of the gesture, effectively cancelling any response to the gesture. In some examples, the passcode itself can denote the gesture.

(18) The ATM **134** may send or receive communication with the computing device **132** over one or more communication networks **140**. In some examples, the ATM **134** can notify the computing device **132** of the input from the user **112** and indicate a location of the ATM **134**.

(19) At block **208**, the process **200** involves verifying that at least one gesture in the input is a distress signal. For example, the process involves determining intent of the gesture from the sensing device. The gesture can have a predetermined meaning. For example, clenching fists can indicate an intent. In some examples, the intent of the gesture can be a distress signal from the user **112**.

(20) At block **210**, the process **200** involves controlling the ATM operation based on the input from the user **112** and the gesture received from the user **112** at the sensing device. For example, when the input and gesture from the user **112** indicates that the user **112** is in distress, the computing device **132** can alter how the an ATM transaction occurs. For example, the computing device **132** can cause the ATM **134** to lockdown the bank account of the user **112**. The ATM **134** can temporarily freeze assets associated with the bank account. The computing device **132** can command the ATM **134** and the sensing device **130** to record audio and video. If a sensing device has multiple cameras, the sensing device **130** can record video from each of the multiple cameras. In some examples, the computing device **132** can instruct the ATM **134** to post a message on a display of the ATM **134** indicating that there are insufficient funds for the transaction such that a suspicious person is aware that the user is walking away without money from the ATM **134**.

(21) At block **212**, the process **200** involves transmitting a request separate from the ATM operation based on the notification of the gesture. The request can be sent from the computing device **132** to authorities **136**. Authorities **136** can include a bank manager, police station, paramedics, an emergency call center, etc., or any suitable combination of the foregoing. In some examples, like when the intention of the gesture indicates distress, the computing device **132** can transmit the request to the authorities **136** that are closest to the location of the user **112** and the ATM **134**.

(22) The computing device **132** can identify a police station located near the ATM **134**. The computing device **132** can contact the police station and report the distress signal at the ATM **134**. The computing device **132** can also inform a bank manager of a bank associated with the ATM **134** of the distress signal. In the request, the computing device **132** can notify the authorities of the intent of the gesture received from the user **112**. In some examples, the computing device **132** can request that the authorities **136** send police officers, paramedics, etc. to the location of the user **112** and the ATM **134**.

(23) FIG. 3 is a block diagram of an example of a computing device **132** for providing advanced security based on gesture recognition according to one example of the present disclosure. The components in FIG. 3, such as a processor **302**, a memory **304**, a bus **306**, and the like, may be integrated into a single structure such as within a single housing of the computing device **132** or

within the ATM **134**. Alternatively, the components shown in FIG. **3** can be distributed from one another and in electrical communication with each other.

(24) As shown, the computing device **132** includes the processor **302** communicatively coupled to the memory **304** by the bus **306**. The processor **302** can include one processor or multiple processors. Non-limiting examples of the processor **302** include a Field-Programmable Gate Array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or any combination of these. The processor **302** can execute instructions **308** stored in the memory **304** to perform operations. In some examples, the instructions **308** stored in the memory **304** can include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C#, or Java.

(25) The memory **304** can include one memory device or multiple memory devices. The memory **304** can be non-volatile and may include any type of memory device that retains stored information when powered off. Non-limiting examples of the memory **304** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any type of non-volatile memory. At least some of the memory **304** can include a non-transitory computer-readable medium from which the processor **302** can read the instructions **308**. The non-transitory computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **302** with the instructions **308** or other program code. Non-limiting examples of the non-transitory computer-readable medium include magnetic disk(s), memory chip(s), RAM, an ASIC, or any other medium from which a computer processor can read instructions.

(26) The computing device **132** also includes the bus **306** (e.g., PCI, ISA, PCI-Express, Hyper-Transport® bus, InfiniBand® bus, NuBus, etc.) and a communications interface **320** (e.g., a Fiber Channel interface, wireless interface, etc.)

(27) Realizations may include fewer or additional components not illustrated in FIG. **3** (e.g., video cards, audio cards, additional communication interfaces, peripheral devices, etc.) The processor **302** and the communication interface **320** are coupled to the bus **306**. Although illustrated as being coupled to the bus **306**, the memory **304** may be coupled to the processor **302**.

(28) Additionally, the memory **304** can include a notification of a gesture **310**, input from a user **312**, intent of the gesture **314**, an ATM operation **316**, and a request **318**. The computing device **132** can receive the notification of the gesture **310** and the intent of the gesture **314** from a sensing device **130** by means of the communications interface **320**. The computing device **132** may send or receive communication with the sensing device **130** and an ATM **134** over one or more communication networks **140**. The input from the user **112** may be received by the computing device **132** from the ATM **134**. The processor **302** may control the ATM operation **316** based on the input from the user **312**. The processor may transmit the request **318** to authorities **136** based on the intent of the gesture **314**.

(29) In some examples, the computing device **132** can implement the process **200** shown in FIG. **2** for effectuating some aspects of the present disclosure. Other examples can involve more operations, fewer operations, different operations, or a different order of the operations shown in FIG. **2**.

(30) The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: a sensing device that comprises a smart device positionable to detect a set of contactless gestures from a user; a processor; and a non-transitory computer-readable medium

that includes instructions executable by the processor to perform operations comprising: receiving a notification of at least one gesture of the set of contactless gestures from the smart device; receiving an input from the user to select an automated teller machine (ATM) operation; determining that the at least one gesture from the smart device is a facial expression that indicates a predetermined meaning defined in the system, wherein the predetermined meaning is a distress signal; verifying that the at least one gesture is the distress signal based on receiving a different and subsequent confirmation input provided by the user, wherein the different and subsequent confirmation input is a second gesture of the set of contactless gestures that is different than the at least one gesture; altering the ATM operation based on the predetermined meaning of the at least one gesture and causing the smart device to record video; identifying a police station located near the ATM; contacting the police station to report the distress signal at the ATM; and informing a bank manager of a bank associated with the ATM of the distress signal.

2. The system of claim 1, wherein the operations further comprise transmitting a request separate from the ATM operation based on the predetermined meaning of the gesture.

3. The system of claim 1, wherein the smart device comprises a smart watch.

4. The system of claim 1, wherein the input from the user comprises a passcode associated with the at least one gesture to confirm the predetermined meaning of the distress signal.

5. The system of claim 1, wherein the at least one gesture is a first facial expression, and wherein the second gesture is a second facial expression that is different from the first facial expression.

6. A computer-implemented method comprising: receiving a notification of at least one gesture from a smart device included with a sensing device of a system, the sensing device configured for detecting a set of contactless gestures that includes the at least one gesture; receiving an input from a user to select an automated teller machine (ATM) operation; determining that the at least one gesture from the smart device is a facial expression that indicates a predetermined meaning defined in a system, wherein the predetermined meaning is a distress signal; verifying that the at least one gesture is the distress signal based on receiving a different and subsequent confirmation input provided by the user, wherein the different and subsequent confirmation input is a second gesture of the set of contactless gestures that is different than the at least one gesture; altering the ATM operation based on the predetermined meaning of the at least one gesture and causing the smart device to record video; identifying a police station located near the ATM; contacting the police station to report the distress signal at the ATM; and informing a bank manager of a bank associated with the ATM of the distress signal.

7. The computer-implemented method of claim 6, further comprising transmitting a request separate from the ATM operation based on the predetermined meaning of the gesture.

8. The computer-implemented method of claim 6, wherein the smart device comprises a smart watch.

9. The computer-implemented method of claim 6, further comprising receiving, by the ATM, a passcode associated with the at least one gesture to confirm the predetermined meaning of the distress signal.

10. A non-transitory computer-readable medium comprising instructions that are executable by a processor for causing the processor to perform operations comprising: receiving a notification of at least one gesture from a smart device included with a sensing device, the sensing device configured for detecting a set of contactless gestures that includes the at least one gesture; receiving an input from a user to select an automated teller machine (ATM) operation; determining that the at least one gesture from the smart device is a facial expression that indicates a predetermined meaning defined in a system, wherein the predetermined meaning is a distress signal; verifying that the at least one gesture is the distress signal based on receiving a different and subsequent confirmation input provided by the user, wherein the different and subsequent confirmation input is a second gesture of the set of contactless gestures that is different than the at least one gesture; altering the ATM operation based on the predetermined meaning of the at least one gesture and causing the

smart device to record video; identifying a police station located near the ATM; contacting the police station to report the distress signal at the ATM; and informing a bank manager of a bank associated with the ATM of the distress signal.

11. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise transmitting a request separate from the ATM operation based on the predetermined meaning of the gesture.

12. The non-transitory computer-readable medium of claim 10, wherein the smart device comprises a smart watch.

13. The non-transitory computer-readable medium of claim 10, wherein the input from the user comprises a passcode associated with the at least one gesture to confirm the predetermined meaning of the distress signal.
